DD-B-583
AUGUST 6, 1958

FEDERAL SPECIFICATION

BOTTLE, INFUSION, KELLY, 700cc.

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers a Kelly design infusion bottle of not less than 700 cubic centimeter (cc.) capacity.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

PPP–B–621—Boxes, Wood, Nailed and Lock-Corner.

PPP–B–636—Boxes, Fiber.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Test Method Std. No. 626—Glassware; Methods for Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL–L–10547 — Liners, Case, Waterproof.

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129 — Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Material.**—Material shall be heavy wall, colorless, borosilicate glass.

3.1.1 *Linear coefficient of expansion.*—The linear coefficient of expansion in the range of 0° or room temperature to 300° Centigrade (C.) shall not exceed 59×10^{-7} centimeter/centimeter °C when tested as specified in 4.3.1.

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3.1.2 *Solubility.*—Glass shall not require more than 1.0 milliliter (ml.) of 0.020*N* sulfuric acid per 10 grams (gm.) of sample, when tested as specified in 4.3.2.

3.1.3 *Thermal shock resistance.* — Bottles shall be subjected to the thermal shock resistance test specified in 4.3.3. Any breakage shall be cause for rejection.

3.1.4 *Annealing.*—The bottles when examined with a polariscope shall not show any ring strain or severe longitudinal strain. A moderate amount of diffused longitudinal strain is acceptable.

3.2 **Construction.**—The bottom of the bottle shall be elongated concentrically to form a tapering tubelike outlet tip, suitable for secure adaption of rubber tubing of ¼-inch inside diameter. The opening shall be not less than 3 millimeters (mm.) in diameter. The top of bottle shall be provided with a small neck 30 mm. (plus or minus 2 mm.) in height, having a uniform outside diameter of from 30 to 36 mm., and an inside diameter from 23 to 25 mm. A raised collar shall be provided to completely encircle the outer periphery of the neck and provide a space between the collar and base of neck of not less than ½-inch for attachment of the hanger. The collar shall be of adequate size to permit secure attachment of the hanger and suspension of the filled bottle. The bottle shall have a capacity of not less than 700 ml.

3.3 **Graduated scale.**—The bottle shall have a suitable engraved, etched, or fused on graduated scale from 0 to 700 ml., reading downward in 25 ml. graduations. The lines for each 100 ml. graduation shall be from 1 to 2 inches in length, and the 25 ml. graduations shall be from ½ to 1½ inches in length, except that the 100 ml. graduation lines shall be longer than the 25 ml. graduation lines. The scale shall be numbered to indicate the 0, 50, 100, 200, 300, 400, 500, 600, and 700 ml. intervals, except that the number "50" may be omitted. All markings shall be clean-cut, legible, permanent, and highly resistant to service wear.

3.3.1 *Permanency of marking.*—The marking shall not discolor the solution and the bottles shall not lose their marking to an extent sufficient to make them less easily readable after being subjected to the tests specified in 4.3.4.

3.4 **Identification marking.** — The bottle shall be clearly and legibly marked with the manufacturer's name or registered trademark.

3.5 **Workmanship.**—Workmanship shall be first class throughout. Infusion bottles shall be reasonably free from seeds, blisters, striae, bubbles or any defects which detract from their appearance and impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Sampling.**

4.1.1 *For visual examination.*—Sampling procedures for visual examination shall be conducted in accordance with the procedures set forth in Military Standard MIL–STD–105.

4.1.2 *For tests.*

4.1.2.1 *For thermal shock test, strain examination, and permanency of marking tests.* —Sample size for each test shall be as indicated for inspection level L–2 in Appendix to Military Standard MIL–STD–105, except that the acceptance number shall be zero (0) for all sample sizes, and a minimum of two samples shall be taken regardless of lot size. The unit of product shall be one bottle.

4.2 **Inspection.**—The inpector shall ascertain that the infusion bottles have been examined and tested in accordance with specification requirements.

4.3 **Tests.**

4.3.1 *Linear coefficient of expansion.*—The glass shall be tested for linear coefficient of

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expansion in accordance with method 240 or 241 of Federal Test Method Standard No. 626.

4.3.2 *Solubility test.*—Bottles shall be subjected to the solubility test in accordance with method 320 of Federal Test Method Standard No. 626.

4.3.3 *Thermal shock test.*—Bottles shall be subjected to the thermal shock test in accordance with method 401 of Federal Test Method Standard No. 626. The bottles shall be heated to a temperature of 100°C.

4.3.4 *Permanency of marking tests.*—In performing the following, different samples shall be used for each test.

4.3.4.1 *Alkali test.*—Immerse samples completely in normal sodium hydroxide at room temperature, cover, and let stand 1½ hours. Remove samples and rinse thoroughly in distilled water. Dry by rubbing vigorously with a cloth.

4.3.4.2 *Acid test (hydrochloric).*—Immerse samples completely in concentrated hydrochloric acid (specific gravity 1.18) at room temperature, and let stand 1½ hours. Remove samples and rinse thoroughly in distilled water. Dry by rubbing vigorously with a cloth.

4.3.4.3 *Acid test (sulphuric).* — Immerse samples completely in concentrated sulphuric acid at room temperature. Let stand 1 hour. Remove samples and rinse thoroughly in distilled water. Dry by rubbing vigorously with a cloth.

4.4 Supplier responsibilities for inspection.—The supplier shall perform examinations and tests set forth in this specification to substantiate conformance of supplies to specification requirements. The examinations and tests shall be at the expense of the supplier. Suppliers shall have adequate test facilities or shall arrange for test facilities acceptable to the Government to be used for executing specific prescribed tests. Records of examinations and tests performed by the supplier shall be maintained by the supplier and made available to the Government, upon the Government's request, at any time, or from time to time, for a period of 2 years after the delivery of the supplies to which these records relate. The supplier shall be responsible for compliance with all requirements of this specification, and all other technical requirements of the contract.

5. PREPARATION FOR DELIVERY

The levels of preservation, packaging, and packing protection used are defined in Federal Standard No. 102 (for civil agencies only).

5.1 Packaging.

5.1.1 *Unit of issue.*—One bottle, as specified, constitutes one unit.

5.1.2 *Packaging quantities.*—The number of units of issue indicated in table I shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

5.1.2.1 *Packing variation permitted.* — If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such

TABLE I.—*Packaging quantities*

Federal stock number	Unit package	Intermediate package	Shipping container
6515–311–9360	1 unit	6 units	36 units

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units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.3 *Level A.*

5.1.3.1 *Unit package.*—Each unit shall be packaged in a solid fiberboard or double-faced corrugated fiberboard carton, of appropriate size and design, constructed in accordance with Federal Specification PPP–B–636. Carton shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of, or damage to the contents. Each carton shall be adequately secured to prevent accidental opening.

5.1.3.2 *Intermediate package.*—Intermediate package shall be either a carton of appropriate size and design, or a complete wrapping with kraft paper of suitable weight. Carton or wrapping shall be adequately secured to prevent accidental opening.

5.1.4 *Level C.*—Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.2 **Packing.**

5.2.1 *Level A.*

5.2.1.1 *Exterior container.*—Exterior container shall be constructed in accordance with Federal Specification PPP–B–621, table IV.

5.2.1.2 *Case liner.*—Each level A box shall be lined with a waterproof liner conforming to Military Specification MIL–L–10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.2.1.3 *Strapping.*—Strapping, when required, shall be in accordance with applicable box specification.

Note—Strapping will not be required for shipment forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 *Level B.*

5.2.2.1 *Exterior container.*—Exterior container shall be constructed in accordance with Federal Specification PPP–B–621, table II.

5.2.3 *Level C.* — The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 **Marking.**

5.3.1 *Unit package.*—Each unit package shall be marked with stock number, item identification, unit, quantity, and name of contractor.

5.3.2 *Intermediate package.*—Each intermediate package shall be marked in accordance with section 5 of Military Standard MIL–STD–129.

5.3.3 *Shipping container.*—Shipping container shall be marked in accordance with Military Standard MIL–STD–129. Service color and service symbol markings are required only on direct shipments from the contractor's plant to overseas activities.

6. NOTES

6.1 Ordering data.—Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.2 This specification covers the following item appearing in the Armed Services Medical Stock List:

6515–311–9350 BOTTLE, INFUSION, Kelly, 700cc.

6.3 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Federal Government requirements.

6.4 **Transportation description.**—Transportation applicable to this item is:

Bottle, glass, NOIBN (other than cut) 1 gallon or less capacity.
Carload minimum weight 30,000 pounds.
Motor volume weight 30,000 pounds.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—M
Navy—Md
Air Force.

Copies of this specification may be purchased for 5 cents each.

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